

MULTI-LEVEL-INVERTER-ARCHITEKTUREN

(MULTI-LEVEL INVERTER ARCHITECTURES)

Introduction

To offer modern electric vehicles cost-effectively, weight, packaging space, and electrical efficiency must be optimized. Multi-Level Inverter technology offers significant potential to optimize the powertrain by enabling more compact, lightweight, and efficient integration into the vehicle. At the same time, the modular system simplifies repair activities and increases the potential for second-life use of the battery system.

Description

TWT develops a model library ranging from the battery cell and power-electronics components to the complete MLI battery system in order to evaluate different architectural approaches through simulation with respect to efficiency and modulation quality. The interaction with the electric machine within the powertrain as well as system behavior during AC and DC charging are investigated.

Timeline: 2025-Present

Technologies/Tools:
MATLAB, Simulink,
Simscape

Goal

By developing a detailed simulation environment, innovative system architectures for automotive traction applications based on Multi-Level Inverter technology can be designed and evaluated. The objective is to increase efficiency and modulation quality while reducing costs and increasing vehicle range. MLI systems can additionally enable reductions in weight and packaging space as well as improvements in maintainability.

TWT-Contribution

Through many years of development experience with MLI architectures in industrial and research projects, TWT has extensive know-how in MLI technology. Combined with expertise in cross-domain modeling and simulation, TWT accelerates the development process toward technology implementation.

Keywords: Virtual Development, Simulation, MLI, Multi-Level Inverter, Battery, Power Electronics, Powertrain, MATLAB, Simulink, Simscape